

WEEK-2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

Task to be Done

1. **The Catalog Search (JOINS):** Create a third table called Loans that connects `member_id` to `book_id`. Write a query to show the Member Name and the Book Title they have currently borrowed.
 2. **Collection Stats (Aggregations):** Write a query to find the total number of books published in each year. o Hint: Use `COUNT(book_id)` and `GROUP BY published_year`.
 3. **The Donation Transaction (ACID):** Use a Transaction (`BEGIN...COMMIT`) to add a new book to the Books table and simultaneously log the entry in a `Donation_History` table.
 4. **Search Speed(Indexing):** Create an Index on the `isbn` column. Explain why searching by ISBN is now faster for the librarian than searching by Title.
- Skill Set Gained**
- Relational Logic: Connecting independent tables (Books + Members) through a Junction table(Loans).
 - Analytical SQL: Using `GROUP BY` to generate business insights.
 - Atomic Operations: Using Transactions to ensure multiple database change succeed or fail as a single unit.

SOLOTIONS:

```
mysql> create database bookflow_db4;
Query OK, 1 row affected (0.01 sec)

mysql> use bookflow_db4;
Database changed
mysql> create table books(book_id int primary key,title varchar(100) not null,isbn varchar(20) unique,published_yr int check(published_yr<2027));
Query OK, 0 rows affected (0.09 sec)

mysql> insert into books(book_id,title,isbn,published_yr)values(1,'The Great Gatsby','9780743273565',1925),(2, 'To Kill a Mockingbird','9780061120084', 1960),(3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from books;
+-----+-----+-----+-----+
| book_id | title | isbn | published_yr |
+-----+-----+-----+-----+
| 1 | The Great Gatsby | 9780743273565 | 1925 |
| 2 | To Kill a Mockingbird | 9780061120084 | 1960 |
| 3 | 1984 | 9780451524935 | 1949 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> create table members(member_id int primary key,full_name varchar(100),email varchar(100) unique);
Query OK, 0 rows affected (0.08 sec)

mysql> Query OK, 0 rows affected (0.08 sec)
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Query OK, 0 rows affected (0.08 sec)' at line 1
mysql> create table members(member_id int primary key,full_name varchar(100),email varchar(100) unique);
ERROR 1050 (42S01): Table 'members' already exists
mysql> drop table members;
Query OK, 0 rows affected (0.06 sec)

mysql> create table members(member_id int primary key,full_name varchar(100),email varchar(100) unique);
```

```
mysql> create table members(member_id int primary key,full_name varchar(100),email varchar(100) unique);
Query OK, 0 rows affected (0.04 sec)

mysql> insert into members(member_id,full_name,email) values (101, 'John Smith', 'john.smith@email.com'),(102, 'Emma Wilson', 'emma.wilson@email.com'),(103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> create table loans(loan_id int primary key,member_id int,book_id int,loan_date date,foreign key(member_id) references members(member_id),foreign key(book_id) references books(book_id));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into loans(loan_id, member_id, book_id, loan_date)values (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.06 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> select * from loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
+-----+-----+-----+-----+
```

```
Command Prompt - mysql -u x + v

mysql> select * from loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT m.full_name AS member_name, b.title AS book_title FROM loans l INNER JOIN members m ON l.member_id = m.member_id INNER JOIN books b ON l.book_id = b.book_id;
+-----+-----+
| member_name | book_title |
+-----+-----+
| John Smith | The Great Gatsby |
| John Smith | To Kill a Mockingbird |
| John Smith | 1984 |
| John Smith | The Great Gatsby |
| Emma Wilson | To Kill a Mockingbird |
| Emma Wilson | The Great Gatsby |
| Emma Wilson | 1984 |
| Michael Brown | 1984 |
| Michael Brown | To Kill a Mockingbird |
| Michael Brown | The Great Gatsby |
+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT published_yr, COUNT(book_id) AS total_books FROM Books GROUP BY published_yr ORDER BY published_yr;
```

```
Command Prompt - mysql -u x + v

mysql> SELECT published_yr, COUNT(book_id) AS total_books FROM Books GROUP BY published_yr ORDER BY published_yr;
+-----+-----+
| published_yr | total_books |
+-----+-----+
| 1925 | 1 |
| 1949 | 1 |
| 1960 | 1 |
+-----+-----+
3 rows in set (0.05 sec)

mysql> CREATE TABLE Donation_History(donation_id int primary key,book_id int,donor_name varchar(100),donation_date DATE,FOREIGN KEY (book_id) REFERENCES books(book_id));
Query OK, 0 rows affected (0.08 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO books (book_id, title, isbn, published_yr) values (4,'Animal Farm','9780451526342',1945);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.05 sec)

mysql> commit;
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE INDEX idx_books_isbn ON books(isbn);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from books where isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn | published_yr |
+-----+-----+-----+-----+
| 3 | 1984 | 9780451524935 | 1949 |
+-----+-----+-----+-----+
```

```
Command Prompt - mysql -u x + v
+-----+
| book_id | title | isbn | published_yr |
+-----+
| 3 | 1984 | 9780451524935 | 1949 |
+-----+
1 row in set (0.00 sec)

mysql> show tables;
+-----+
| Tables_in_bookflow_db4 |
+-----+
| books |
| donation_history |
| loans |
| members |
+-----+
4 rows in set (0.05 sec)

mysql> select * from books;
+-----+
| book_id | title | isbn | published_yr |
+-----+
| 1 | The Great Gatsby | 9780743273565 | 1925 |
| 2 | To Kill a Mockingbird | 9780061120084 | 1960 |
| 3 | 1984 | 9780451524935 | 1949 |
| 4 | Animal Farm | 9780451526342 | 1945 |
+-----+
4 rows in set (0.00 sec)

mysql> select * from donation_history;
+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+
| 1 | 4 | Raj Kumar | 2026-07-30 |
+-----+
1 row in set (0.05 sec)
```

```
Command Prompt - mysql -u x + v

mysql> select * from donation_history;
+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+
| 1 | 4 | Raj Kumar | 2026-07-30 |
+-----+
1 row in set (0.05 sec)

mysql> select * from loans;
+-----+
| loan_id | member_id | book_id | loan_date |
+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+
10 rows in set (0.00 sec)

mysql> select * from members;
+-----+
| member_id | full_name | email |
+-----+
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+
3 rows in set (0.04 sec)

mysql> |
```

Problem Statement 1: SkyTrack Airline Management System

A regional airline company currently manages its flight schedules, passenger records, and ticket bookings using spreadsheets. As the number of flights and passengers continues to grow, it has become difficult to track reservations, maintain accurate records, and generate operational reports. To improve efficiency, the airline plans to develop a database system called SkyTrack. As a database developer, your task is to create a database named `skytrack_db` and design three tables: Flights, Passengers, and Bookings. The Flights table should contain details such as `flight_id`, `flight_number`, `source`, `destination`, `departure_date`, and `ticket_price`, while the Passengers table should store `passenger_id`, `passenger_name`, and `email`. The Bookings table should connect passengers and flights using `passenger_id` and `flight_id` as foreign keys. Apply appropriate constraints, including PRIMARY KEY, UNIQUE, and CHECK constraints to ensure data integrity. Insert at least 10 records into each table. Display all table contents and write an SQL query using INNER JOIN to show the passenger name, flight number, source, and destination for every booking. Generate a report showing the total number of flights operating to each destination using COUNT() and GROUP BY. Create a Flight_History table and demonstrate the use of a Transaction by adding a new flight and simultaneously recording the entry in the history table, ensuring both operations succeed or fail together. Finally, create an Index on the `flight_number` column and explain why searching by flight number is faster than searching by destination. This exercise develops skills in relational database design, JOIN operations, aggregate functions, transactions, and indexing.

Solutions:

```
Command Prompt - mysql -u × + v

mysql> create database skytrack_db;
Query OK, 1 row affected (0.01 sec)

mysql> use skytrack_db;
Database changed
mysql> CREATE TABLE Flights (
    ->     flight_id INT PRIMARY KEY,
    ->     flight_number VARCHAR(10) UNIQUE NOT NULL,
    ->     source VARCHAR(50) NOT NULL,
    ->     destination VARCHAR(50) NOT NULL,
    ->     departure_date DATE NOT NULL,
    ->     ticket_price DECIMAL(10,2) CHECK(ticket_price > 0)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE TABLE Passengers (
    ->     passenger_id INT PRIMARY KEY,
    ->     passenger_name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE Bookings (
    ->     booking_id INT PRIMARY KEY,
    ->     passenger_id INT,
    ->     flight_id INT,
    ->     booking_date DATE,
    ->
    ->     FOREIGN KEY(passenger_id) REFERENCES Passengers(passenger_id),
    ->
    ->     FOREIGN KEY(flight_id) REFERENCES Flights(flight_id)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Flights VALUES
    -> (101, 'AI101', 'Hyderabad', 'Delhi', '2026-08-10', 4500),
```

```
Command Prompt - mysql -u X + v
-> (102,'AI102','Delhi','Mumbai','2026-08-11',3800),
-> (103,'AI103','Chennai','Bangalore','2026-08-12',2200),
-> (104,'AI104','Mumbai','Hyderabad','2026-08-13',3400),
-> (105,'AI105','Kolkata','Delhi','2026-08-14',4200),
-> (106,'AI106','Hyderabad','Goa','2026-08-15',3000),
-> (107,'AI107','Delhi','Chennai','2026-08-16',5000),
-> (108,'AI108','Goa','Mumbai','2026-08-17',2700),
-> (109,'AI109','Bangalore','Hyderabad','2026-08-18',2600),
-> (110,'AI110','Pune','Delhi','2026-08-19',3900);
Query OK, 10 rows affected (0.05 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Passengers VALUES
-> (1,'Rahul Sharma','rahul@gmail.com'),
-> (2,'Priya Reddy','priya@gmail.com'),
-> (3,'Arjun Kumar','arjun@gmail.com'),
-> (4,'Sneha Patel','sneha@gmail.com'),
-> (5,'Vikram Rao','vikram@gmail.com'),
-> (6,'Anjali Singh','anjali@gmail.com'),
-> (7,'Kiran Das','kiran@gmail.com'),
-> (8,'Neha Verma','neha@gmail.com'),
-> (9,'Rohan Gupta','rohan@gmail.com'),
-> (10,'Pooja Mehta','pooja@gmail.com');
Query OK, 10 rows affected (0.05 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Bookings VALUES
-> (1001,1,101,'2026-07-30'),
-> (1002,2,102,'2026-07-30'),
-> (1003,3,103,'2026-07-30'),
-> (1004,4,104,'2026-07-31'),
-> (1005,5,105,'2026-07-31'),
-> (1006,6,106,'2026-07-31'),
-> (1007,7,107,'2026-08-01'),
-> (1008,8,108,'2026-08-01'),
-> (1009,9,109,'2026-08-01'),
-> (1010,10,110,'2026-08-02');
```

```
Command Prompt - mysql -u X + v
Query OK, 10 rows affected (0.05 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Flights;
+-----+-----+-----+-----+-----+-----+
| flight_id | flight_number | source | destination | departure_date | ticket_price |
+-----+-----+-----+-----+-----+-----+
| 101 | AI101 | Hyderabad | Delhi | 2026-08-10 | 4500.00 |
| 102 | AI102 | Delhi | Mumbai | 2026-08-11 | 3800.00 |
| 103 | AI103 | Chennai | Bangalore | 2026-08-12 | 2200.00 |
| 104 | AI104 | Mumbai | Hyderabad | 2026-08-13 | 3400.00 |
| 105 | AI105 | Kolkata | Delhi | 2026-08-14 | 4200.00 |
| 106 | AI106 | Hyderabad | Goa | 2026-08-15 | 3000.00 |
| 107 | AI107 | Delhi | Chennai | 2026-08-16 | 5000.00 |
| 108 | AI108 | Goa | Mumbai | 2026-08-17 | 2700.00 |
| 109 | AI109 | Bangalore | Hyderabad | 2026-08-18 | 2600.00 |
| 110 | AI110 | Pune | Delhi | 2026-08-19 | 3900.00 |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM Passengers;
+-----+-----+-----+
| passenger_id | passenger_name | email |
+-----+-----+-----+
| 1 | Rahul Sharma | rahul@gmail.com |
| 2 | Priya Reddy | priya@gmail.com |
| 3 | Arjun Kumar | arjun@gmail.com |
| 4 | Sneha Patel | sneha@gmail.com |
| 5 | Vikram Rao | vikram@gmail.com |
| 6 | Anjali Singh | anjali@gmail.com |
| 7 | Kiran Das | kiran@gmail.com |
| 8 | Neha Verma | neha@gmail.com |
| 9 | Rohan Gupta | rohan@gmail.com |
| 10 | Pooja Mehta | pooja@gmail.com |
+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u X + v
mysql> SELECT * FROM Bookings;
+-----+-----+-----+-----+
| booking_id | passenger_id | flight_id | booking_date |
+-----+-----+-----+-----+
| 1001 | 1 | 101 | 2026-07-30 |
| 1002 | 2 | 102 | 2026-07-30 |
| 1003 | 3 | 103 | 2026-07-30 |
| 1004 | 4 | 104 | 2026-07-31 |
| 1005 | 5 | 105 | 2026-07-31 |
| 1006 | 6 | 106 | 2026-07-31 |
| 1007 | 7 | 107 | 2026-08-01 |
| 1008 | 8 | 108 | 2026-08-01 |
| 1009 | 9 | 109 | 2026-08-01 |
| 1010 | 10 | 110 | 2026-08-02 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT
-> P.passenger_name,
-> F.flight_number,
-> F.source,
-> F.destination
-> FROM Bookings B
-> INNER JOIN Passengers P
-> ON B.passenger_id=P.passenger_id
-> INNER JOIN Flights F
-> ON B.flight_id=F.flight_id;
+-----+-----+-----+-----+
| passenger_name | flight_number | source | destination |
+-----+-----+-----+-----+
| Rahul Sharma | AI101 | Hyderabad | Delhi |
| Priya Reddy | AI102 | Delhi | Mumbai |
| Arjun Kumar | AI103 | Chennai | Bangalore |
| Sneha Patel | AI104 | Mumbai | Hyderabad |
| Vikram Rao | AI105 | Kolkata | Delhi |
| Anjali Singh | AI106 | Hyderabad | Goa |
| Kiran Das | AI107 | Delhi | Chennai |
+-----+-----+-----+-----+
```

```
Command Prompt - mysql -u X + v
+-----+-----+-----+-----+
| Neha Verma | AI108 | Goa | Mumbai |
| Rohan Gupta | AI109 | Bangalore | Hyderabad |
| Pooja Mehta | AI110 | Pune | Delhi |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT
-> destination,
-> COUNT(*) AS Total_Flights
-> FROM Flights
-> GROUP BY destination;
+-----+-----+
| destination | Total_Flights |
+-----+-----+
| Delhi | 3 |
| Mumbai | 2 |
| Bangalore | 1 |
| Hyderabad | 2 |
| Goa | 1 |
| Chennai | 1 |
+-----+-----+
6 rows in set (0.05 sec)

mysql> CREATE TABLE Flight_History(
-> history_id INT PRIMARY KEY,
-> flight_id INT,
-> flight_number VARCHAR(10),
-> action_performed VARCHAR(30),
-> action_date DATETIME
-> );
Query OK, 0 rows affected (0.07 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Flights
```

```
Command Prompt - mysql -u X + v

mysql> INSERT INTO Flights
-> VALUES
-> (111,'AI111','Hyderabad','Pune','2026-08-20',3600);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Flight_History
-> VALUES
-> (1,111,'AI111','Flight Added',NOW());
Query OK, 1 row affected (0.04 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_flight_number
-> ON Flights(flight_number);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT *
-> FROM Flights
-> WHERE flight_number='AI105';
+-----+-----+-----+-----+-----+-----+
| flight_id | flight_number | source | destination | departure_date | ticket_price |
+-----+-----+-----+-----+-----+-----+
| 105 | AI105 | Kolkata | Delhi | 2026-08-14 | 4200.00 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM Flights
-> WHERE destination='Delhi';
+-----+-----+-----+-----+-----+-----+
| flight_id | flight_number | source | destination | departure_date | ticket_price |
+-----+-----+-----+-----+-----+-----+
| 101 | AI101 | Hyderabad | Delhi | 2026-08-10 | 4500.00 |
| 105 | AI105 | Kolkata | Delhi | 2026-08-14 | 4200.00 |
| 110 | AI110 | Pune | Delhi | 2026-08-19 | 3900.00 |
+-----+-----+-----+-----+-----+-----+
```

```
Command Prompt - mysql -u X + v

+-----+-----+-----+-----+-----+-----+
| flight_id | flight_number | source | destination | departure_date | ticket_price |
+-----+-----+-----+-----+-----+-----+
| 101 | AI101 | Hyderabad | Delhi | 2026-08-10 | 4500.00 |
| 105 | AI105 | Kolkata | Delhi | 2026-08-14 | 4200.00 |
| 110 | AI110 | Pune | Delhi | 2026-08-19 | 3900.00 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> show tables;
+-----+
| Tables_in_skytrack_db |
+-----+
| bookings |
| flight_history |
| flights |
| passengers |
+-----+
4 rows in set (0.05 sec)

mysql> select * from bookings;
+-----+-----+-----+-----+
| booking_id | passenger_id | flight_id | booking_date |
+-----+-----+-----+-----+
| 1001 | 1 | 101 | 2026-07-30 |
| 1002 | 2 | 102 | 2026-07-30 |
| 1003 | 3 | 103 | 2026-07-30 |
| 1004 | 4 | 104 | 2026-07-31 |
| 1005 | 5 | 105 | 2026-07-31 |
| 1006 | 6 | 106 | 2026-07-31 |
| 1007 | 7 | 107 | 2026-08-01 |
| 1008 | 8 | 108 | 2026-08-01 |
| 1009 | 9 | 109 | 2026-08-01 |
| 1010 | 10 | 110 | 2026-08-02 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
Command Prompt - mysql -u < >
mysql> select * from flight_history;
+-----+-----+-----+-----+-----+
| history_id | flight_id | flight_number | action_performed | action_date |
+-----+-----+-----+-----+-----+
| 1 | 111 | AI111 | Flight Added | 2026-07-30 10:20:59 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> select * from flights;
+-----+-----+-----+-----+-----+-----+
| flight_id | flight_number | source | destination | departure_date | ticket_price |
+-----+-----+-----+-----+-----+-----+
| 101 | AI101 | Hyderabad | Delhi | 2026-08-10 | 4500.00 |
| 102 | AI102 | Delhi | Mumbai | 2026-08-11 | 3800.00 |
| 103 | AI103 | Chennai | Bangalore | 2026-08-12 | 2200.00 |
| 104 | AI104 | Mumbai | Hyderabad | 2026-08-13 | 3400.00 |
| 105 | AI105 | Kolkata | Delhi | 2026-08-14 | 4200.00 |
| 106 | AI106 | Hyderabad | Goa | 2026-08-15 | 3000.00 |
| 107 | AI107 | Delhi | Chennai | 2026-08-16 | 5000.00 |
| 108 | AI108 | Goa | Mumbai | 2026-08-17 | 2700.00 |
| 109 | AI109 | Bangalore | Hyderabad | 2026-08-18 | 2600.00 |
| 110 | AI110 | Pune | Delhi | 2026-08-19 | 3900.00 |
| 111 | AI111 | Hyderabad | Pune | 2026-08-20 | 3600.00 |
+-----+-----+-----+-----+-----+-----+
11 rows in set (0.00 sec)

mysql> select * from passengers;
+-----+-----+-----+
| passenger_id | passenger_name | email |
+-----+-----+-----+
| 1 | Rahul Sharma | rahul@gmail.com |
| 2 | Priya Reddy | priya@gmail.com |
| 3 | Arjun Kumar | arjun@gmail.com |
| 4 | Sneha Patel | sneha@gmail.com |
| 5 | Vikram Rao | vikram@gmail.com |
| 6 | Anjali Singh | anjali@gmail.com |
| 7 | Kiran Das | kiran@gmail.com |
+-----+-----+-----+
```

```
Command Prompt - mysql -u < >
mysql> select * from flights;
+-----+-----+-----+-----+-----+-----+
| flight_id | flight_number | source | destination | departure_date | ticket_price |
+-----+-----+-----+-----+-----+-----+
| 101 | AI101 | Hyderabad | Delhi | 2026-08-10 | 4500.00 |
| 102 | AI102 | Delhi | Mumbai | 2026-08-11 | 3800.00 |
| 103 | AI103 | Chennai | Bangalore | 2026-08-12 | 2200.00 |
| 104 | AI104 | Mumbai | Hyderabad | 2026-08-13 | 3400.00 |
| 105 | AI105 | Kolkata | Delhi | 2026-08-14 | 4200.00 |
| 106 | AI106 | Hyderabad | Goa | 2026-08-15 | 3000.00 |
| 107 | AI107 | Delhi | Chennai | 2026-08-16 | 5000.00 |
| 108 | AI108 | Goa | Mumbai | 2026-08-17 | 2700.00 |
| 109 | AI109 | Bangalore | Hyderabad | 2026-08-18 | 2600.00 |
| 110 | AI110 | Pune | Delhi | 2026-08-19 | 3900.00 |
| 111 | AI111 | Hyderabad | Pune | 2026-08-20 | 3600.00 |
+-----+-----+-----+-----+-----+-----+
11 rows in set (0.00 sec)

mysql> select * from passengers;
+-----+-----+-----+
| passenger_id | passenger_name | email |
+-----+-----+-----+
| 1 | Rahul Sharma | rahul@gmail.com |
| 2 | Priya Reddy | priya@gmail.com |
| 3 | Arjun Kumar | arjun@gmail.com |
| 4 | Sneha Patel | sneha@gmail.com |
| 5 | Vikram Rao | vikram@gmail.com |
| 6 | Anjali Singh | anjali@gmail.com |
| 7 | Kiran Das | kiran@gmail.com |
| 8 | Neha Verma | neha@gmail.com |
| 9 | Rohan Gupta | rohan@gmail.com |
| 10 | Pooja Mehta | pooja@gmail.com |
+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> |
```

Problem Statement 2: MediCare Hospital Management System

A hospital currently maintains patient records, doctor information, and appointment schedules manually, which often results in scheduling conflicts, misplaced records, and delays in accessing important information. To modernize its operations, the hospital wants to develop a database system called MediCare HMS. As a database developer, your task is to create a database named `medicare_db` and design three tables: Doctors, Patients, and Appointments. The Doctors table should store details such as `doctor_id`, `doctor_name`, specialization, and `consultation_fee`, while the Patients table should contain `patient_id`, `patient_name`, and email. The Appointments table should connect doctors and patients using foreign keys and include an appointment date. Apply appropriate constraints such as PRIMARY KEY, UNIQUE, and CHECK constraints to ensure data accuracy and consistency. Insert at least 10 records into each table. Display all table contents and write an SQL query using INNER JOIN to show the patient name, doctor name, specialization, and appointment date for all appointments. Generate a report displaying the number of doctors available in each specialization using COUNT() and GROUP BY. Create a Doctor_History table and demonstrate the use of a Transaction by registering a new doctor and simultaneously recording the registration in the history table, ensuring both operations are completed successfully as a single unit. Finally, create an Index on the specialization column and explain how indexing improves search performance when locating doctors by specialization. This exercise helps students understand database design, table relationships, SQL queries, aggregate functions, transactions, and query optimization techniques.

SOLUTIONS:

```
mysql> create database medicare_db;
Query OK, 1 row affected (0.05 sec)

mysql> use medicare_db;
Database changed
mysql> CREATE TABLE Doctors (
  ->     doctor_id INT PRIMARY KEY,
  ->     doctor_name VARCHAR(100) NOT NULL,
  ->     specialization VARCHAR(50) NOT NULL,
  ->     consultation_fee DECIMAL(10,2) CHECK (consultation_fee > 0)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Patients (
  ->     patient_id INT PRIMARY KEY,
  ->     patient_name VARCHAR(100) NOT NULL,
  ->     email VARCHAR(100) UNIQUE NOT NULL
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Appointments (
  ->     appointment_id INT PRIMARY KEY,
  ->     doctor_id INT,
  ->     patient_id INT,
  ->     appointment_date DATE,
  ->
  ->     FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),
  ->     FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Doctors VALUES
  -> (101,'Dr. Rajesh Kumar','Cardiology',800),
  -> (102,'Dr. Anitha Reddy','Neurology',1000),
  -> (103,'Dr. Vivek Sharma','Orthopedics',700),
  -> (104,'Dr. Priya Singh','Dermatology',600),
  -> (105,'Dr. Arjun Patel','Pediatrics',500),
```

```

Command Prompt - mysql -u X + v
-> (105,'Dr. Arjun Patel','Pediatrics',500),
-> (106,'Dr. Sneha Rao','Gynecology',900),
-> (107,'Dr. Kiran Das','ENT',550),
-> (108,'Dr. Meera Joshi','Ophthalmology',650),
-> (109,'Dr. Rohit Verma','General Medicine',400),
-> (110,'Dr. Kavya Nair','Cardiology',850);

```

Query OK, 10 rows affected (0.05 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Patients VALUES
```

```

-> (1,'Rahul Sharma','rahul@gmail.com'),
-> (2,'Priya Reddy','priya@gmail.com'),
-> (3,'Arjun Kumar','arjun@gmail.com'),
-> (4,'Sneha Patel','sneha@gmail.com'),
-> (5,'Vikram Rao','vikram@gmail.com'),
-> (6,'Anjali Singh','anjali@gmail.com'),
-> (7,'Kiran Gupta','kiran@gmail.com'),
-> (8,'Neha Verma','neha@gmail.com'),
-> (9,'Rohan Das','rohan@gmail.com'),
-> (10,'Pooja Mehta','pooja@gmail.com');

```

Query OK, 10 rows affected (0.05 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> INSERT INTO Appointments VALUES
```

```

-> (1001,101,1,'2026-08-01'),
-> (1002,102,2,'2026-08-02'),
-> (1003,103,3,'2026-08-03'),
-> (1004,104,4,'2026-08-04'),
-> (1005,105,5,'2026-08-05'),
-> (1006,106,6,'2026-08-06'),
-> (1007,107,7,'2026-08-07'),
-> (1008,108,8,'2026-08-08'),
-> (1009,109,9,'2026-08-09'),
-> (1010,110,10,'2026-08-10');

```

Query OK, 10 rows affected (0.01 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Doctors;
```

doctor_id	doctor_name	specialization	consultation_fee
101	Dr. Rajesh Kumar	Cardiology	800.00
102	Dr. Anitha Reddy	Neurology	1000.00
103	Dr. Vivek Sharma	Orthopedics	700.00
104	Dr. Priya Singh	Dermatology	600.00
105	Dr. Arjun Patel	Pediatrics	500.00
106	Dr. Sneha Rao	Gynecology	900.00
107	Dr. Kiran Das	ENT	550.00
108	Dr. Meera Joshi	Ophthalmology	650.00
109	Dr. Rohit Verma	General Medicine	400.00
110	Dr. Kavya Nair	Cardiology	850.00

10 rows in set (0.00 sec)

```
mysql> SELECT * FROM Patients;
```

patient_id	patient_name	email
1	Rahul Sharma	rahul@gmail.com
2	Priya Reddy	priya@gmail.com
3	Arjun Kumar	arjun@gmail.com
4	Sneha Patel	sneha@gmail.com
5	Vikram Rao	vikram@gmail.com
6	Anjali Singh	anjali@gmail.com
7	Kiran Gupta	kiran@gmail.com
8	Neha Verma	neha@gmail.com
9	Rohan Das	rohan@gmail.com
10	Pooja Mehta	pooja@gmail.com

10 rows in set (0.00 sec)

```
Command Prompt - mysql -u × + v
mysql> SELECT * FROM Appointments;
+-----+-----+-----+-----+
| appointment_id | doctor_id | patient_id | appointment_date |
+-----+-----+-----+-----+
| 1001 | 101 | 1 | 2026-08-01 |
| 1002 | 102 | 2 | 2026-08-02 |
| 1003 | 103 | 3 | 2026-08-03 |
| 1004 | 104 | 4 | 2026-08-04 |
| 1005 | 105 | 5 | 2026-08-05 |
| 1006 | 106 | 6 | 2026-08-06 |
| 1007 | 107 | 7 | 2026-08-07 |
| 1008 | 108 | 8 | 2026-08-08 |
| 1009 | 109 | 9 | 2026-08-09 |
| 1010 | 110 | 10 | 2026-08-10 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT
-> P.patient_name,
-> D.doctor_name,
-> D.specialization,
-> A.appointment_date
-> FROM Appointments A
-> INNER JOIN Patients P
-> ON A.patient_id = P.patient_id
-> INNER JOIN Doctors D
-> ON A.doctor_id = D.doctor_id;
+-----+-----+-----+-----+
| patient_name | doctor_name | specialization | appointment_date |
+-----+-----+-----+-----+
| Rahul Sharma | Dr. Rajesh Kumar | Cardiology | 2026-08-01 |
| Priya Reddy | Dr. Anitha Reddy | Neurology | 2026-08-02 |
| Arjun Kumar | Dr. Vivek Sharma | Orthopedics | 2026-08-03 |
| Sneha Patel | Dr. Priya Singh | Dermatology | 2026-08-04 |
| Vikram Rao | Dr. Arjun Patel | Pediatrics | 2026-08-05 |
| Anjali Singh | Dr. Sneha Rao | Gynecology | 2026-08-06 |
| Kiran Gupta | Dr. Kiran Das | ENT | 2026-08-07 |
+-----+-----+-----+-----+
```

```
Command Prompt - mysql -u × + v
| Vikram Rao | Dr. Arjun Patel | Pediatrics | 2026-08-05 |
| Anjali Singh | Dr. Sneha Rao | Gynecology | 2026-08-06 |
| Kiran Gupta | Dr. Kiran Das | ENT | 2026-08-07 |
| Neha Verma | Dr. Meera Joshi | Ophthalmology | 2026-08-08 |
| Rohan Das | Dr. Rohit Verma | General Medicine | 2026-08-09 |
| Pooja Mehta | Dr. Kavya Nair | Cardiology | 2026-08-10 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT
-> specialization,
-> COUNT(*) AS Total_Doctors
-> FROM Doctors
-> GROUP BY specialization;
+-----+-----+
| specialization | Total_Doctors |
+-----+-----+
| Cardiology | 2 |
| Neurology | 1 |
| Orthopedics | 1 |
| Dermatology | 1 |
| Pediatrics | 1 |
| Gynecology | 1 |
| ENT | 1 |
| Ophthalmology | 1 |
| General Medicine | 1 |
+-----+-----+
9 rows in set (0.00 sec)

mysql> CREATE TABLE Doctor_History (
-> history_id INT PRIMARY KEY,
-> doctor_id INT,
-> doctor_name VARCHAR(100),
-> specialization VARCHAR(50),
-> action_performed VARCHAR(50),
-> action_date DATETIME
-> );
```

```
Command Prompt - mysql -u x + v
-> action_date DATETIME
-> );
Query OK, 0 rows affected (0.07 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Doctors
-> VALUES
-> (111,'Dr. Suresh Reddy','Psychiatry',750);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Doctor_History
-> VALUES
-> (1,111,'Dr. Suresh Reddy','Psychiatry','Doctor Registered',NOW());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_specialization
-> ON Doctors(specialization);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> CREATE INDEX idx_specialization
-> ON Doctors(specialization);
ERROR 1061 (42000): Duplicate key name 'idx_specialization'
mysql> SELECT *
-> FROM Doctors
-> WHERE specialization = 'Cardiology';
```

```
Command Prompt - mysql -u x + v
-> WHERE specialization = 'Cardiology';
+-----+-----+-----+-----+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----+-----+-----+-----+
| 101 | Dr. Rajesh Kumar | Cardiology | 800.00 |
| 110 | Dr. Kavya Nair | Cardiology | 850.00 |
+-----+-----+-----+-----+
2 rows in set (0.05 sec)

mysql> show tables;
+-----+
| Tables_in_medicare_db |
+-----+
| appointments |
| doctor_history |
| doctors |
| patients |
+-----+
4 rows in set (0.00 sec)

mysql> select * from appointments;
+-----+-----+-----+-----+
| appointment_id | doctor_id | patient_id | appointment_date |
+-----+-----+-----+-----+
| 1001 | 101 | 1 | 2026-08-01 |
| 1002 | 102 | 2 | 2026-08-02 |
| 1003 | 103 | 3 | 2026-08-03 |
| 1004 | 104 | 4 | 2026-08-04 |
| 1005 | 105 | 5 | 2026-08-05 |
| 1006 | 106 | 6 | 2026-08-06 |
| 1007 | 107 | 7 | 2026-08-07 |
| 1008 | 108 | 8 | 2026-08-08 |
| 1009 | 109 | 9 | 2026-08-09 |
| 1010 | 110 | 10 | 2026-08-10 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select * from doctor_history;
```

history_id	doctor_id	doctor_name	specialization	action_performed	action_date
1	111	Dr. Suresh Reddy	Psychiatry	Doctor Registered	2026-07-30 11:19:03

```
1 row in set (0.00 sec)
```



```
mysql> select * from doctors;
```

doctor_id	doctor_name	specialization	consultation_fee
101	Dr. Rajesh Kumar	Cardiology	800.00
102	Dr. Anitha Reddy	Neurology	1000.00
103	Dr. Vivek Sharma	Orthopedics	700.00
104	Dr. Priya Singh	Dermatology	600.00
105	Dr. Arjun Patel	Pediatrics	500.00
106	Dr. Sneha Rao	Gynecology	900.00
107	Dr. Kiran Das	ENT	550.00
108	Dr. Meera Joshi	Ophthalmology	650.00
109	Dr. Rohit Verma	General Medicine	400.00
110	Dr. Kavya Nair	Cardiology	850.00
111	Dr. Suresh Reddy	Psychiatry	750.00

```
11 rows in set (0.00 sec)
```



```
mysql> select * from patients;
```

patient_id	patient_name	email
1	Rahul Sharma	rahul@gmail.com
2	Priya Reddy	priya@gmail.com
3	Arjun Kumar	arjun@gmail.com
4	Sneha Patel	sneha@gmail.com
5	Vikram Rao	vikram@gmail.com
6	Anjali Singh	anjali@gmail.com

```
mysql> select * from doctors;
```

doctor_id	doctor_name	specialization	consultation_fee
101	Dr. Rajesh Kumar	Cardiology	800.00
102	Dr. Anitha Reddy	Neurology	1000.00
103	Dr. Vivek Sharma	Orthopedics	700.00
104	Dr. Priya Singh	Dermatology	600.00
105	Dr. Arjun Patel	Pediatrics	500.00
106	Dr. Sneha Rao	Gynecology	900.00
107	Dr. Kiran Das	ENT	550.00
108	Dr. Meera Joshi	Ophthalmology	650.00
109	Dr. Rohit Verma	General Medicine	400.00
110	Dr. Kavya Nair	Cardiology	850.00
111	Dr. Suresh Reddy	Psychiatry	750.00

```
11 rows in set (0.00 sec)
```



```
mysql> select * from patients;
```

patient_id	patient_name	email
1	Rahul Sharma	rahul@gmail.com
2	Priya Reddy	priya@gmail.com
3	Arjun Kumar	arjun@gmail.com
4	Sneha Patel	sneha@gmail.com
5	Vikram Rao	vikram@gmail.com
6	Anjali Singh	anjali@gmail.com
7	Kiran Gupta	kiran@gmail.com
8	Neha Verma	neha@gmail.com
9	Rohan Das	rohan@gmail.com
10	Pooja Mehta	pooja@gmail.com

```
10 rows in set (0.00 sec)
```



```
mysql> |
```